

Representation-Level Uniqueness for the $n - k = 3$ Layer of Nonnormal Cayley Graphs on Symmetric Groups

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Abstract

For $n \geq 5$, $1 \leq r < n - 3$, and $H = C(n, n - 3; r)$, let $\rho_\lambda(H) = \sum_{h \in H} \rho_\lambda(h)$ be the adjacency block corresponding to the irreducible representation of S_n indexed by $\lambda \vdash n$. We prove the $n - k = 3$ layer of Conjecture 4.7 of Li, Xia, and Zhou. If n is odd, the second largest eigenvalue is attained only by the standard representation $(n - 1, 1)$. If n is even, the strictly second largest eigenvalue is attained exactly by $(n - 1, 1)$ and its sign twist $(2, 1^{n-2})$. In either case the target value is

$$\mu_{n,r} = \frac{(n-5)!(n-r-2)}{6} (2n^2 - nr - r^2 - 8n + 5r),$$

with multiplicity $n - r - 1$ inside each attaining block. For $n \geq 7$ the proof is symbolic: an endpoint equality audit reduces to two near-standard representations, explicit two-subset and exterior-square models give a zero-intersection argument, and strictness is propagated from $r + 1$ to r using only the known non-strict $n - k = 2$ bound. The three remaining cases $n = 5, 6$ are certified by exact characteristic polynomials.

1 Introduction

For integers $1 \leq r < k < n$, let $C(n, k; r)$ be the set of k -cycles in S_n whose support contains $[r] = \{1, \dots, r\}$. These inverse-closed sets define generally nonnormal Cayley graphs. Simons and Zaleski introduced this family and computed the spectrum of the natural permutation block, including the value now denoted by $\mu_2(n, k; r)$ [SZ22]. Li, Xia, and Zhou proved that this value is the second, or strictly second, eigenvalue for $k \leq n - 2$, but their recursive argument gives only a non-strict upper bound on the other irreducible blocks [LXZ26, Theorems 4.4 and 4.6]. Their Conjecture 4.7 asks for the corresponding representation-level uniqueness [LXZ26, Conjecture 4.7].

This note treats the complete infinite slice $k = n - 3$. The distinction between the value problem and the uniqueness problem is essential. The known value theorem supplies an inequality $\lambda_{\max} \rho_\lambda(H) \leq \mu_2$ for the relevant non-degree blocks. The result below makes that inequality strict except at the stated representations. The proof has three components.

First, at the endpoint $r = n - 4$, we enlarge the setting to a three-step family H_q with $q = 2, 3, 4$. Equality in the Weyl bound forces a common vector in a standard child module and in the invariant space of a long-cycle class. Branching leaves only $(N - 2, 2)$ and $(N - 2, 1, 1)$, together with their sign twists when appropriate. A two-subset model and an exterior-square model show that both intersections vanish. Second, for $r < n - 4$, the decomposition of $C(n, n - 3; r)$ according to whether a chosen point is moved combines a non-strict $n - k = 2$ estimate with the already strict estimate at $r + 1$. Thus strictness propagates backwards without assuming the still stronger uniqueness statement at $n - k = 2$. Third, exact finite certificates handle the only cases not covered by the symbolic argument, namely $(n, k, r) = (5, 2, 1), (6, 3, 1), (6, 3, 2)$.

The result proved here is only the $n - k = 3$ layer; no assertion is made for $n - k \neq 3$. Nor does this note assert public priority. The literature discussion records the sources used by the proof but does not turn an absence in a finite search into a claim of novelty.

2 Definitions, representation blocks, and the main result

Let G be a finite group and let $S = S^{-1} \subseteq G \setminus \{1\}$. The Cayley graph $\text{Cay}(G, S)$ has vertex set G and edges $\{\{g, gs\} : g \in G, s \in S\}$. Its adjacency operator is right multiplication by $S^+ = \sum_{s \in S} s$ on the regular representation. If $\widehat{G}$ is a complete set of irreducible representations, the regular representation decomposition gives

$$A(\text{Cay}(G, S)) \sim \bigoplus_{\rho \in \widehat{G}} (\dim \rho) \rho(S), \quad \rho(S) = \sum_{s \in S} \rho(s). \quad (2.1)$$

Here $(\dim \rho) \rho(S)$ means $\dim \rho$ copies of the block. In particular, a block eigenvalue of multiplicity a contributes $a \dim \rho$ to its multiplicity in the whole graph [LXZ26, Proposition 2.1].

Irreducible representations of S_n are indexed by partitions $\lambda \vdash n$; we write S^λ and ρ_λ for the corresponding Specht module and representation. The trivial, sign, and standard representations correspond respectively to (n) , (1^n) , and $(n-1, 1)$. Conjugating the Young diagram twists by the sign character:

$$S^{\lambda'} \cong S^\lambda \otimes \text{sgn}. \quad (2.2)$$

The branching rule is multiplicity-free:

$$S^\lambda \downarrow_{S_{n-1}} \cong \bigoplus_{\nu \nearrow \lambda} S^\nu, \quad (2.3)$$

where ν runs over partitions obtained by deleting one removable box from λ [LXZ26, Theorem 2.5].

Fix now

$$n \geq 5, \quad k = n - 3, \quad 1 \leq r \leq n - 4,$$

and put

$$H_{n,r} = C(n, n-3; r) = \{\sigma \in S_n : \sigma \text{ is an } (n-3)\text{-cycle and } [r] \subseteq \text{supp}(\sigma)\}.$$

Writing $m = n - r$, direct counting gives

$$d_{n,r} := |H_{n,r}| = \binom{m}{3} (n-4)!. \quad (2.4)$$

Because $H_{n,r}$ is inverse-closed, every block

$$T_\lambda(n, r) := \rho_\lambda(H_{n,r}) = \sum_{h \in H_{n,r}} \rho_\lambda(h) \quad (2.5)$$

is self-adjoint in a unitary realization. We denote its largest eigenvalue by $\Lambda_\lambda(n, r)$.

The sign of a k -cycle is $(-1)^{k-1}$. Thus when n is odd, $k = n - 3$ is even, the generators are odd, and $H_{n,r}$ generates S_n . When n is even, k is odd, the generators are even, and $H_{n,r}$ generates A_n ; consequently the graph on S_n has two isomorphic components. These generation statements are the specialization of [LXZ26, Lemma 2.14]. It follows that the degree occurs only in the trivial block for odd n , and in the trivial and sign blocks for even n .

Define

$$\begin{aligned} \mu_{n,r} &= \frac{(n-5)!(m-2)}{6} (3(n-4)(m-1) - (m-4)(m-3)) \\ &= \frac{(n-5)!(n-r-2)}{6} (2n^2 - nr - r^2 - 8n + 5r). \end{aligned} \quad (2.6)$$

Theorem 2.1 (The $n - k = 3$ layer of Conjecture 4.7). *Let $n \geq 5$ and $1 \leq r \leq n - 4$.*

1. *If n is odd, then*

$$\Lambda_{(n-1,1)}(n, r) = \mu_{n,r},$$

the eigenvalue $\mu_{n,r}$ has multiplicity $n - r - 1$ inside the standard block, and

$$\Lambda_\lambda(n, r) < \mu_{n,r} \quad \text{for every } \lambda \notin \{(n), (n-1, 1)\}.$$

Hence the second largest graph eigenvalue is attained only by the standard representation.

2. *If n is even, then the trivial and sign blocks both have eigenvalue $d_{n,r}$, while*

$$\Lambda_{(n-1,1)}(n, r) = \Lambda_{(2,1^{n-2})}(n, r) = \mu_{n,r}.$$

The target has multiplicity $n - r - 1$ inside each of these two blocks, and

$$\Lambda_\lambda(n, r) < \mu_{n,r} \quad \text{for every } \lambda \notin \{(n), (1^n), (n-1, 1), (2, 1^{n-2})\}.$$

Hence the strictly second largest graph eigenvalue is attained exactly by the standard representation and its sign twist.

In the regular representation the multiplicity of $\mu_{n,r}$ is therefore $(n-1)(n-r-1)$ for odd n and $2(n-1)(n-r-1)$ for even n .

3 The standard block

We begin with a direct computation of the target value and its multiplicity. This also fixes the normalization used in the rest of the paper.

Lemma 3.1 (Standard-block calculation). *The largest eigenvalue of $T_{(n-1,1)}(n, r)$ is $\mu_{n,r}$, with multiplicity exactly $m - 1 = n - r - 1$. If n is even, the sign-twisted standard block has the same spectrum.*

Proof. Let $\mathcal{A} = [r]$ and $\mathcal{B} = [n] \setminus [r]$, so $|\mathcal{B}| = m$. On the natural permutation module with basis $e_1, \dots, e_n$, let P_h denote the permutation matrix of h and put $M = \sum_{h \in H_{n,r}} P_h$. Conjugation invariance under $S_{\mathcal{A}} \times S_{\mathcal{B}}$ shows that the entries of M depend only on the parts containing their row and column indices and on whether the indices agree. Set

$$a = (n-5)! \binom{m}{3}, \quad b = (n-5)! \binom{m-1}{3}, \quad c = (n-4)! \binom{m-1}{2}, \quad e = (n-5)! \binom{m-2}{3}, \quad (3.1)$$

with the convention $\binom{u}{v} = 0$ when $v < 0$ or $v > u$.

For distinct $i, j \in \mathcal{A}$, prescribing $h(i) = j$ leaves $(n-5)! \binom{m}{3} = a$ possibilities. If exactly one of i, j belongs to $\mathcal{B}$, the analogous number is b . For $i \in \mathcal{B}$, the diagonal entry counts generators fixing i and equals c . For distinct $i, j \in \mathcal{B}$, prescribing $h(i) = j$ gives e possibilities. Thus, on

$$U_{\mathcal{B}} = \{x \in \mathbb{R}^n : \text{supp}(x) \subseteq \mathcal{B}, \sum_{i \in \mathcal{B}} x_i = 0\},$$

the operator M acts by $c - e$. This space has dimension $m - 1$, is orthogonal to the constant vector, and hence lies in the standard summand of the natural module.

The remaining invariant spaces are the constant line, the space

$$U_{\mathcal{A}} = \{x : \text{supp}(x) \subseteq \mathcal{A}, \sum_{i \in \mathcal{A}} x_i = 0\},$$

of dimension $r - 1$, and the one-dimensional space of vectors constant on each of $\mathcal{A}$ and $\mathcal{B}$ and orthogonal to the constant vector. Their eigenvalues are respectively $d_{n,r}$, $-a$, and $(r - 1)a - rb$. Moreover,

$$c - e > 0 > -a$$

and a Pascal-identity simplification gives

$$\frac{(c - e) - ((r - 1)a - rb)}{(n - 5)!} = (m - 3) \binom{m - 1}{2} + \binom{m - 2}{2} > 0. \quad (3.2)$$

The four invariant spaces have total dimension $1 + (m - 1) + (r - 1) + 1 = n$, so $c - e$ is the largest standard-block eigenvalue and has multiplicity exactly $m - 1$. Finally,

$$\begin{aligned} c - e &= (n - 4)! \binom{m - 1}{2} - (n - 5)! \binom{m - 2}{3} \\ &= \frac{(n - 5)!(m - 2)}{6} (3(n - 4)(m - 1) - (m - 4)(m - 3)) = \mu_{n,r}. \end{aligned}$$

If n is even, every generator is even; (2.2) therefore makes the standard and conjugate-standard operators identical. $\square$

4 Endpoint strictness

Put $p = n - 4$. The endpoint $r = n - 4$ will be obtained from an auxiliary family on $p + q$ letters. Fix a set R of size p and define

$$H_q = C(p + q, p + 1; p), \quad q \in \{2, 3, 4\}, \quad (4.1)$$

where the notation means that every $(p + 1)$ -cycle moves all points of R . The standard-block formula of Siemons and Zalesski, or direct substitution in (2.6) in its general form, gives

$$|H_q| = q p!, \quad \mu(H_q) = (q - 1)p!. \quad (4.2)$$

We use two previously proved inputs from Li–Xia–Zhou.

Proposition 4.1 (Published spectral inputs). *The following statements hold.*

1. *Let $N \geq 6$ and $k = N - 2$. If k is even, every irreducible block other than the trivial degree block has largest eigenvalue at most $\mu_2(N, k; r)$. If k is odd, the same bound holds after excluding the trivial and sign degree blocks. No uniqueness assertion is included in this input.*
2. *Let $N = p + 2$ and $r = p = N - 2$, with $p \geq 3$. If $p + 1$ is even, the value $p!$ is attained only by the standard representation. If $p + 1$ is odd, it is attained exactly by the standard representation and its sign twist.*

Proof. Part (1) is precisely the blockwise non-strict consequence of [LXZ26, Theorems 4.4 and 4.6]. It is weaker than Conjecture 4.7. For part (2), when $p + 1$ is even, N is odd and [LXZ26, Theorem 3.3(d)] applies at $r = N - 2$; when $p + 1$ is odd, N is even and [LXZ26, Theorem 3.5(b.2)] applies. $\square$

Proposition 4.2 (Endpoint family). *Let $p \geq 3$ and $q \in \{2, 3, 4\}$. Apart from the degree block or blocks, the value $(q - 1)p!$ for H_q is attained only by the standard representation if $p + 1$ is even, and exactly by the standard representation and its sign twist if $p + 1$ is odd.*

Proof of Proposition 4.2: reduction. We induct on q . The case $q = 2$ is Proposition 4.1(2). Suppose $q \in \{3, 4\}$ and the assertion is known for $q - 1$. Write $N = p + q$, choose $t \notin R$, and set $Y = R \cup \{t\}$, so $|Y| = p + 1$. Partition H_q according to whether a cycle moves t :

$$H_q = H_{q-1} \dot{\cup} K_Y, \quad (4.3)$$

where the first term is embedded in the point stabilizer S_{N-1} and K_Y is the full conjugacy class of $(p + 1)$ -cycles in S_Y . On an irreducible S_N -module write the corresponding operators as

$$T = A + B.$$

The operator B is a sum of $p!$ unitary matrices, hence

$$\lambda_{\max}(B) \leq p!. \quad (4.4)$$

If a unit vector v gives equality, then $\operatorname{Re}\langle v, \rho(g)v \rangle \leq 1$ for every $g \in K_Y$, and equality in the sum forces $\rho(g)v = v$ for every such g . The class K_Y generates S_Y when $p + 1$ is even and A_Y when $p + 1$ is odd. Thus a top vector for B is respectively S_Y - or A_Y -invariant.

Now let S^λ be neither a degree module nor one of the target parent modules in the statement. By branching, its restriction to S_{N-1} contains no degree module. In the odd-cycle case it contains neither the trivial nor the sign module. The induction hypothesis therefore gives

$$\lambda_{\max}(A) \leq (q - 2)p!. \quad (4.5)$$

If equality holds, its vector lies in a standard child module, with a conjugate-standard child also allowed when $p + 1$ is odd. Weyl's inequality now yields $\lambda_{\max}(T) \leq (q - 1)p!$. Equality in this inequality would force one and the same vector to be a top vector for both A and B .

Adding one box to the child partition $(N - 2, 1)$ and removing the target parent $(N - 1, 1)$ leaves only

$$(N - 2, 2), \quad (N - 2, 1, 1), \quad (4.6)$$

and, in the odd-cycle case, their conjugates. The next two lemmas show that the required common vector does not exist.

Lemma 4.3 (The two-subset zero intersection). *In $S^{(N-2,2)}$, the standard S_{N-1} -child on which A can attain $(q - 2)p!$ has zero intersection with the top eigenspace of B .*

Proof. The two-subset permutation module decomposes as

$$\mathbb{C} \binom{[N]}{2} \cong S^{(N)} \oplus S^{(N-1,1)} \oplus S^{(N-2,2)}.$$

Realize the last summand as symmetric arrays $x_{ij} = x_{ji}$, $x_{ii} = 0$, satisfying

$$\sum_{j \neq i} x_{ij} = 0 \quad (1 \leq i \leq N). \quad (4.7)$$

Relabel so that $t = N$. The unique standard child under the point stabilizer S_{N-1} consists of the arrays

$$x_{ij} = u_i + u_j \quad (i, j < N), \quad x_{it} = -(N - 3)u_i, \quad (4.8)$$

where $\sum_{i < N} u_i = 0$. These arrays satisfy (4.7), form an $(N - 2)$ -dimensional standard module, and hence are exactly the possible equality space for A .

Suppose first that an array (4.8) is S_Y -invariant. Since $p \geq 3$, take distinct $a, b \in R$. Transitivity of S_Y on the two-subsets of Y and comparison of x_{ab} with x_{at} show, after varying a, b , that all u_a for $a \in R$ equal a common value c . Then

$$2c = -(N - 3)c,$$

so $c = 0$. If $o \notin Y$, invariance on the pairs meeting o gives

$$u_o = x_{ao} = x_{to} = -(N-3)u_o,$$

and hence $u_o = 0$. Thus the intersection with the S_Y -invariants is zero.

It remains to treat the case in which $p+1$ is odd and a top vector for B is only known to be A_Y -invariant. Here p is even and $p \geq 4$, so $|Y| \geq 5$. Restricted to S_Y , the two-subset permutation module is

$$\mathbb{C} \binom{Y}{2} \oplus (q-1)\mathbb{C}Y \oplus \binom{q-1}{2} \mathbf{1}. \quad (4.9)$$

None of these permutation modules contains the sign representation of S_Y : by Frobenius reciprocity, the sign character has no fixed vector under a two-subset or one-subset stabilizer, each of which contains an odd permutation. Since $A_Y \triangleleft S_Y$ has index two, the A_Y -invariant subspace of any S_Y -module is a sum of trivial and sign modules. In (4.9) only the trivial part can occur. Hence the A_Y - and S_Y -invariant subspaces coincide, and the preceding zero-intersection argument applies. $\square$

Lemma 4.4 (The exterior-square zero intersection). *In $S^{(N-2,1,1)}$, the standard S_{N-1} -child on which A can attain $(q-2)p!$ has zero intersection with the top eigenspace of B .*

Proof. Let

$$V_N = \{(z_1, \dots, z_N) \in \mathbb{C}^N : \sum_i z_i = 0\}$$

be the standard module. There is a standard isomorphism $S^{(N-2,1,1)} \cong \bigwedge^2 V_N$. With $t = N$, put

$$w = (N-1)e_t - \sum_{i < N} e_i.$$

Under S_{N-1} we have $V_N = V_{N-1} \oplus \mathbb{C}w$, and the unique standard child in the exterior square is

$$W = \{u \wedge w : u \in V_{N-1}\}. \quad (4.10)$$

Regard u as having $u_t = 0$ and $\sum_{i < N} u_i = 0$. The antisymmetric coefficients of $u \wedge w$ are

$$c_{ij} = u_j - u_i \quad (i, j < t), \quad c_{it} = (N-1)u_i \quad (i < t), \quad (4.11)$$

with $c_{ji} = -c_{ij}$.

If $u \wedge w$ is S_Y -invariant, a transposition exchanging the two indices of any coefficient inside Y changes its sign, so that coefficient must vanish. In particular $c_{at} = (N-1)u_a = 0$ for every $a \in R$. For $o \notin Y$, invariance under the transposition exchanging a and t gives

$$c_{ao} = c_{to},$$

which by (4.11) becomes $u_o = -(N-1)u_o$. Hence every u_o also vanishes, and (4.10) has zero intersection with the S_Y -invariants.

When $p+1$ is odd, $|Y| \geq 5$ and

$$V_N \downarrow_{S_Y} \cong V_Y \oplus \mathbf{1}^{\oplus(q-1)}.$$

Consequently its exterior square is a direct sum of $\bigwedge^2 V_Y \cong S^{(|Y|-2,1,1)}$, copies of V_Y , and trivial modules. It contains no sign module. The same index-two argument as in Lemma 4.3 therefore identifies the A_Y -invariants with the S_Y -invariants, and the intersection is again zero. $\square$

Completion. The two lemmas rule out equality for the two partitions in (4.6). If $p+1$ is odd, every generator appearing in A and B is an even permutation. Global sign twisting therefore leaves both operators unchanged, so the same zero-intersection conclusions apply to the conjugate partitions. This proves the induction step from $q-1$ to q and hence the proposition for $q = 2, 3, 4$. $\square$

Taking $q = 4$ and $p = n-4$ in Proposition 4.2 proves the required strict classification at $r = n-4$ for every $n \geq 7$.

5 Backward propagation from the endpoint

We now fix $n \geq 7$ and $k = n - 3$. Suppose $1 \leq r \leq n - 5$ and that the strict classification of Theorem 2.1 is already known at $r + 1$. Choose $t \notin [r]$, relabel it as $r + 1$, and split the generators according to whether they move t :

$$C(n, n - 3; r) = C(n - 1, n - 3; r) \dot{\cup} C(n, n - 3; r + 1). \quad (5.1)$$

On a fixed parent module S^λ , write the two corresponding operators as A and B .

Let S^λ be one of the blocks that must be strictly below the target in Theorem 2.1. Its restriction to S_{n-1} contains no trivial child; in the odd- k case it contains no sign child either. Indeed, by branching, a trivial child can occur only for parent (n) or $(n - 1, 1)$, and a sign child only for parent (1^n) or $(2, 1^{n-2})$, all of which have been excluded. Since

$$(n - 1) - (n - 3) = 2,$$

Proposition 4.1(1), applied to every irreducible child, gives

$$\lambda_{\max}(A) \leq \mu_2(n - 1, n - 3; r). \quad (5.2)$$

This is only the published numerical bound, not a uniqueness assertion for the $n - k = 2$ layer.

The operator B is the parameter- $r + 1$ operator on the same parent module. The backward induction hypothesis therefore supplies the strict estimate

$$\lambda_{\max}(B) < \mu_{n, r+1}. \quad (5.3)$$

The standard-block formula satisfies the recurrence

$$\mu_{n, r} = \mu_2(n - 1, n - 3; r) + \mu_{n, r+1}, \quad (5.4)$$

which follows either by direct substitution or from [LXZ26, Equation (10)]. Weyl's inequality now gives

$$\lambda_{\max} T_\lambda(n, r) \leq \lambda_{\max}(A) + \lambda_{\max}(B) < \mu_{n, r}.$$

The strictness comes entirely from the second summand; no equality classification at $n - k = 2$ is needed. Starting from the endpoint $r = n - 4$ and descending through $n - 5, n - 6, \dots, 1$ proves all strict inequalities for $n \geq 7$.

6 The exact low-order certificate

The endpoint argument assumes $p = n - 4 \geq 3$. The only remaining legal triples are

$$(n, k, r) = (5, 2, 1), \quad (6, 3, 1), \quad (6, 3, 2). \quad (6.1)$$

For completeness, these cases were computed exactly in every irreducible block. The computation uses no floating-point arithmetic. For a standard Young tableau T , let $d_i(T) = \text{content}_T(i + 1) - \text{content}_T(i)$. In the Young orthogonal basis the adjacent transposition s_i acts by

$$\rho(s_i)e_T = \frac{1}{d_i(T)}e_T + \sqrt{1 - \frac{1}{d_i(T)^2}}e_{s_i T}, \quad (6.2)$$

where the second term is zero when $s_i T$ is not standard. Enumerating the connection set, expressing each cycle in adjacent transpositions, and summing (6.2) gives an exact symmetric block. A diagonally similar rational seminormal form is used to compute the characteristic polynomial over $\mathbb{Q}[x]$.

The certificate checks all partitions, the hook-length dimensions and $\sum_{\lambda \vdash n} (\dim S^\lambda)^2 = n!$, the Coxeter relations, exact enumeration and inverse closure of the connection set, the standard-block quotient, sign twisting, and the regular-representation trace identities

$$\sum_{\lambda} d_{\lambda} \operatorname{tr} T_{\lambda} = 0, \quad \sum_{\lambda} d_{\lambda} \operatorname{tr}(T_{\lambda}^2) = n!|H|. \quad (6.3)$$

The complete characteristic polynomials for the three cases are listed in Section A. They show that the target is 3 only in $(4, 1)$ for $n = 5$, and is respectively 11 and 6 only in $(5, 1)$ and $(2, 1^4)$ for the two $n = 6$ cases, with the block multiplicities stated in Theorem 2.1.

The larger exact run comprised seven parameter cases and 96 irreducible blocks. Only the 29 blocks in (6.1) are used to discharge the low-order proof obligation. The remaining 67 blocks, for $(7, 4, r)$ with $1 \leq r \leq 3$ and $(8, 5, 4)$, are interface and regression checks; they are not used in place of the symbolic proof for general n .

Proof of Theorem 2.1. Lemma 3.1 establishes the target value and its within-block multiplicity. For $n \geq 7$, Proposition 4.2 proves the strict endpoint classification and Section 5 propagates it to every r . For $n = 5, 6$, the exact polynomials in Section A establish the same classification. The degree-block statement follows from the generation and parity discussion after (2.5), and the whole-graph multiplicities follow from (2.1) because both target representations have dimension $n - 1$. $\square$

7 Related work, scope, and verification status

Siemons and Zalesski studied $C(n, k; r)$ and established the natural-block lower bound that gives $\mu_2(n, k; r)$ [SZ22]. Li, Xia, and Zhou proved the Aldous-property value for $k \leq n - 2$, classified the endpoint $k = n - 1$ in the ranges used here, and formulated the stricter representation-level uniqueness statement as Conjecture 4.7 [LXZ26]. Their recurrence and non-strict block bound are inputs to the backward induction; their endpoint classification is the base $q = 2$ of the endpoint audit.

The present proof neither changes the hypotheses of Conjecture 4.7 nor addresses any layer other than $n - k = 3$. Its general part is independent of the 96-block experiment. The exact computation is logically required only for the three low-order cases in (6.1), and its additional cases serve as checks on formulas, parity, multiplicities, and the proof-computation interface.

The mathematical argument and this final exposition were checked independently. The reviews covered the parity split, representation multiplicities, endpoint equality analysis, backward induction, all exact low-order cases, the source mapping, and every displayed characteristic polynomial in the appendix, and returned the verdict **CORRECT**. Public novelty or priority is **NOT ESTABLISHED**.

A Complete low-order characteristic polynomials

In the following tables, $P_{\lambda}(x) = \det(xI - T_{\lambda}(n, r))$ and m_{μ} is the multiplicity of the target $\mu_{n,r}$ inside the block. Every partition of the relevant integer appears exactly once.

$(n, k, r) = (5, 2, 1)$: $|H| = 4$, $\mu = 3$

λ	d_λ	$P_\lambda(x)$	m_μ
(5)	1	$x - 4$	0
(4, 1)	4	$(x - 3)^3(x + 1)$	3
(3, 2)	5	$x^3(x - 2)^2$	0
(3, 1, 1)	6	$(x - 2)^3(x + 2)^3$	0
(2, 2, 1)	5	$x^3(x + 2)^2$	0
(2, 1, 1, 1)	4	$(x - 1)(x + 3)^3$	0
(1 ⁵)	1	$x + 4$	0

$(n, k, r) = (6, 3, 1)$: $|H| = 20$, $\mu = 11$

λ	d_λ	$P_\lambda(x)$	m_μ
(6)	1	$x - 20$	0
(5, 1)	5	$(x - 11)^4(x + 4)$	4
(4, 2)	9	$(x - 4)^5(x + 5)^4$	0
(4, 1, 1)	10	$(x - 4)^6(x + 1)^4$	0
(3, 3)	5	$(x + 4)^5$	0
(3, 2, 1)	16	$(x + 1)^{10}(x + 5)^6$	0
(3, 1, 1, 1)	10	$(x - 4)^6(x + 1)^4$	0
(2, 2, 2)	5	$(x + 4)^5$	0
(2, 2, 1, 1)	9	$(x - 4)^5(x + 5)^4$	0
(2, 1 ⁴)	5	$(x - 11)^4(x + 4)$	4
(1 ⁶)	1	$x - 20$	0

$(n, k, r) = (6, 3, 2)$: $|H| = 8$, $\mu = 6$

λ	d_λ	$P_\lambda(x)$	m_μ
(6)	1	$x - 8$	0
(5, 1)	5	$(x - 6)^3(x - 2)(x + 4)$	3
(4, 2)	9	$(x - 4)^2(x - 2)^3(x + 2)(x + 4)^3$	0
(4, 1, 1)	10	$x^3(x - 4)^3(x - 2)(x + 2)^3$	0
(3, 3)	5	$x^3(x + 4)^2$	0
(3, 2, 1)	16	$(x - 1)^6(x + 1)^4(x + 3)^6$	0
(3, 1, 1, 1)	10	$x^3(x - 4)^3(x - 2)(x + 2)^3$	0
(2, 2, 2)	5	$x^3(x + 4)^2$	0
(2, 2, 1, 1)	9	$(x - 4)^2(x - 2)^3(x + 2)(x + 4)^3$	0
(2, 1 ⁴)	5	$(x - 6)^3(x - 2)(x + 4)$	3
(1 ⁶)	1	$x - 8$	0

References

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